

Profile

Software engineer and PhD researcher focused on machine learning, distributed computing, and numerical software. Strong background in Python and C++ development for simulation, automation, GPU acceleration, data systems, and infrastructure spanning research and production-adjacent environments.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Developed graph neural networks and automatic-differentiation models in Python to predict material properties including optical emission spectra.
- Developed scientific software for first-principles and many-body simulation, combining Python and C++ with MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and CMake.
- Sharded JAX/PyTorch computational-graph workloads across up to 800 Frontier GPUs and TB-scale datasets, with experience spanning NVIDIA and AMD GPU systems on national-lab clusters.
- Used gdb for stack, frame, and memory inspection alongside cuda-gdb, nvprof, TotalView, and Linaro DDT for MPI/GPU distributed debugging, compiler-aware builds, and performance analysis.
- Extended into ML-assisted and retrieval-driven automation with LangChain, OpenAI API models, and local Ollama models for coding and research setup.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related first-principles work.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built C++/CUDA parallel-computing applications with OpenCL fallback for large acoustic waveform and sensor datasets used in engineering diagnostics and field-tool analysis.
- Built C++ GUI applications with OpenGL custom plotting controls for rendering and interrogating large sensor datasets; used gdb to inspect stacks, frames, and memory during debugging.
- Implemented a supporting C#/ASP.NET web server and Microsoft SQL Server services for remote tool testing, calibration, data collection, and processing.
- Worked close to embedded and instrumentation systems, giving practical experience with software operating alongside deployed hardware and field constraints.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical content for optics and photonics topics, demonstrating strong technical communication across specialized engineering subject matter.

Selected Projects

Machine Learning Models for Material Properties

- Developed PyTorch Geometric graph neural networks and automatic-differentiation models for molecular-property prediction and optical-properties calculations in material systems.

Agentic Coding and Scientific Workflow Generation

- Developed LangChain-based retrieval-driven tooling using OpenAI API models and local Ollama models to read documentation and academic papers, then generate structured starter inputs and workflow artifacts.

Container Orchestration for First-Principles Computing

- Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Matrix-Multiply AI Accelerator

- Developed a Verilog-based matrix-multiply accelerator project for AI-oriented compute pipelines and hardware-software performance exploration.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing work combining scientific computing, high-performance workflows, and computational materials research.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Selected Coursework

Computing Foundations: Theoretical Computer Science; Probabilistic Machine Learning

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, semiconductor physics, and numerical/scientific computing practice

Technical Skills

Languages: Python, C, C++, CUDA, HIP, JAX, PyTorch, C#, Bash, PowerShell

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux command line, CMake, C++ GUI development, OpenGL custom plotting controls, Docker, Docker Compose, Kubernetes

GPU / Debug: CUDA, ROCm/HIP, cuBLAS, cuTENSOR, cuRAND, rocBLAS, rocFFT, gdb, cuda-gdb, nvprof, TotalView, Linaro DDT

Platforms: Frontier, Frontera, Perlmutter, Linux, CMake, Docker

ML / Automation: PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, transformers, agentic automation